Задание 1

```
lly@lly ~/m/e/t/1/webapp-chart (master)> helm lint
==> Linting .
[INFO] Chart.yaml: icon is recommended

1 chart(s) linted, 0 chart(s) failed
lly@lly ~/m/e/t/1/webapp-chart (master)>
```

helm lint

```
lly@lly ~/m/e/t/1/webapp-chart (master)> tree
.
├── charts
├── Chart.yaml
├── templates
│   ├── deployment.yaml
│   ├── _helpers.tpl
│   ├── hpa.yaml
│   ├── httproute.yaml
│   ├── ingress.yaml
│   ├── NOTES.txt
│   ├── serviceaccount.yaml
│   ├── service.yaml
│   └── tests
│       └── test-connection.yaml
└── values.yaml

4 directories, 11 files
```

tree

```
lly@lly ~/m/e/t/1/webapp-chart (master)> helm install webapp-release ./
NAME: webapp-release
LAST DEPLOYED: Sun Apr 12 17:30:00 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace default -l "app.kubernetes.io/nam
e=webapp-chart,app.kubernetes.io/instance=webapp-release" -o jsonpath="{.items[0].
metadata.name}")
  export CONTAINER_PORT=$(kubectl get pod --namespace default $POD_NAME -o jsonpat
h="{.spec.containers[0].ports[0].containerPort}")
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl --namespace default port-forward $POD_NAME 8080:$CONTAINER_PORT
```

helm install webapp-release ./

```
lly@lly ~/m/e/t/1/webapp-chart (master)> helm get manifest webapp-release
---
# Source: webapp-chart/templates/serviceaccount.yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: webapp-release-webapp-chart
  labels:
    helm.sh/chart: webapp-chart-0.1.0
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: webapp-release
    app.kubernetes.io/version: "1.16.0"
    app.kubernetes.io/managed-by: Helm
automountServiceAccountToken: true
---
```

```yaml
---
# Source: webapp-chart/templates/service.yaml
apiVersion: v1
kind: Service
metadata:
  name: webapp-release-webapp-chart
  labels:
    helm.sh/chart: webapp-chart-0.1.0
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: webapp-release
    app.kubernetes.io/version: "1.16.0"
    app.kubernetes.io/managed-by: Helm
    environment: production
    team: backend
spec:
  type: ClusterIP
  ports:
    - port: 8080
      targetPort:
      protocol: TCP
      name: http
  selector:
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: webapp-release
---
```

```yaml
# Source: webapp-chart/templates/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: webapp-release-webapp-chart
  labels:
    helm.sh/chart: webapp-chart-0.1.0
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: webapp-release
    app.kubernetes.io/version: "1.16.0"
    app.kubernetes.io/managed-by: Helm
    environment: production
    team: backend
spec:
  replicas: 3
  selector:
    matchLabels:
      app.kubernetes.io/name: webapp-chart
      app.kubernetes.io/instance: webapp-release
  template:
    metadata:
      labels:
        helm.sh/chart: webapp-chart-0.1.0
        app.kubernetes.io/name: webapp-chart
        app.kubernetes.io/instance: webapp-release
        app.kubernetes.io/version: "1.16.0"
        app.kubernetes.io/managed-by: Helm
```

```
  spec:
    serviceAccountName: webapp-release-webapp-chart
    containers:
      - name: webapp-chart
        image: "nginx:1.23"
        imagePullPolicy: IfNotPresent
        ports:
          - name: http
            containerPort: 80
            protocol: TCP
        livenessProbe:
          httpGet:
            path: /
            port: http
        readinessProbe:
          httpGet:
            path: /
            port: http
        resources:
          limits:
            cpu: 200m
            memory: 256Mi
          requests:
            cpu: 200m
            memory: 256Mi
```

helm get manifest webapp-release

```
lly@lly ~/m/e/t/1/webapp-chart (master)> kubectl get deployments -l environment=pr
oduction
NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
webapp-release-webapp-chart   3/3     3            3           2m7s
```

kubectl get deployments -l environment=production

```
lly@lly ~/m/e/t/1/webapp-chart (master)> kubectl get svc -l team=backend
NAME                        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)
  AGE
webapp-release-webapp-chart   ClusterIP   10.111.165.135   <none>        8080/TCP
  2m43s
```

kubectl get svc -l team=backend

```
lly@lly ~/m/e/t/1/webapp-chart (master)> kubectl describe deployment/webapp-releas
e-webapp-chart
Name:                   webapp-release-webapp-chart
Namespace:              default
CreationTimestamp:      Sun, 12 Apr 2026 17:29:58 +0300
Labels:                 app.kubernetes.io/instance=webapp-release
                        app.kubernetes.io/managed-by=Helm
                        app.kubernetes.io/name=webapp-chart
                        app.kubernetes.io/version=1.16.0
                        environment=production
                        helm.sh/chart=webapp-chart-0.1.0
                        team=backend
Annotations:            deployment.kubernetes.io/revision: 1
                        meta.helm.sh/release-name: webapp-release
                        meta.helm.sh/release-namespace: default
Selector:               app.kubernetes.io/instance=webapp-release,app.kubernetes.i
o/name=webapp-chart
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavaila
ble
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:           app.kubernetes.io/instance=webapp-release
                    app.kubernetes.io/managed-by=Helm
                    app.kubernetes.io/name=webapp-chart
                    app.kubernetes.io/version=1.16.0
                    helm.sh/chart=webapp-chart-0.1.0
  Service Account:  webapp-release-webapp-chart
```

```
  Containers:
   webapp-chart:
    Image:      nginx:1.23
    Port:       80/TCP (http)
    Host Port:  0/TCP (http)
    Limits:
      cpu:     200m
      memory:  256Mi
    Requests:
      cpu:         200m
      memory:      256Mi
    Liveness:      http-get http://:http/ delay=0s timeout=1s period=10s #success=
1 #failure=3
    Readiness:     http-get http://:http/ delay=0s timeout=1s period=10s #success=
1 #failure=3
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   webapp-release-webapp-chart-945c5cd49 (3/3 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  3m22s  deployment-controller  Scaled up replica set w
ebapp-release-webapp-chart-945c5cd49 from 0 to 3
```

kubectl desribe deployment/webapp-release-webapp-chart

Задание 2

```
lly@lly ~/m/e/t/2/webapp-chart (master)> helm template demo-chart ./ --debug
install.go:225: 2026-04-12 18:43:07.227116406 +0300 MSK m=+0.020515453 [debug] Original ch
art version: ""
install.go:242: 2026-04-12 18:43:07.227225996 +0300 MSK m=+0.020625033 [debug] CHART PATH:
 /home/lly/ma/eleventh/tasks/2/webapp-chart

---
# Source: webapp-chart/templates/serviceaccount.yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: demo-chart-webapp-chart
  labels:
    helm.sh/chart: webapp-chart-0.1.0
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: demo-chart
    app.kubernetes.io/version: "1.16.0"
    app.kubernetes.io/managed-by: Helm
automountServiceAccountToken: true
---
```

```
# Source: webapp-chart/templates/service.yaml
apiVersion: v1
kind: Service
metadata:
  name: demo-chart-webapp-chart
  labels:
    helm.sh/chart: webapp-chart-0.1.0
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: demo-chart
    app.kubernetes.io/version: "1.16.0"
    app.kubernetes.io/managed-by: Helm
    environment: production
    team: backend
    course: "kubernetes"
    project: "university"
    semester: "spring"

spec:
  type: ClusterIP
  ports:
    - port: 8080
      targetPort:
      protocol: TCP
      name: http
  selector:
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: demo-chart
---
```

```yaml
# Source: webapp-chart/templates/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: demo-chart-webapp-chart
  labels:
    helm.sh/chart: webapp-chart-0.1.0
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: demo-chart
    app.kubernetes.io/version: "1.16.0"
    app.kubernetes.io/managed-by: Helm
    environment: production
    team: backend
    course: "kubernetes"
    project: "university"
    semester: "spring"
env:
  - name: APP_ENV
    value: production
  - name: APP_DEBUG
    value: false
  - name: LOG_LEVEL
    value: info
```

```yaml
spec:
  replicas: 3
  selector:
    matchLabels:
      app.kubernetes.io/name: webapp-chart
      app.kubernetes.io/instance: demo-chart
  template:
    metadata:
      labels:
        helm.sh/chart: webapp-chart-0.1.0
        app.kubernetes.io/name: webapp-chart
        app.kubernetes.io/instance: demo-chart
        app.kubernetes.io/version: "1.16.0"
        app.kubernetes.io/managed-by: Helm
    spec:
      serviceAccountName: demo-chart-webapp-chart
      containers:
        - name: webapp-chart
          image: "nginx:1.23"
          imagePullPolicy: IfNotPresent
          ports:
            - name: http
              containerPort: 80
              protocol: TCP
          livenessProbe:
            httpGet:
              path: /
              port: http
          readinessProbe:
            httpGet:
              path: /
              port: http
          resources:
            limits:
              cpu: 200m
              memory: 256Mi
```

```
# Source: webapp-chart/templates/tests/test-connection.yaml
apiVersion: v1
kind: Pod
metadata:
  name: "demo-chart-webapp-chart-test-connection"
  labels:
    helm.sh/chart: webapp-chart-0.1.0
    app.kubernetes.io/name: webapp-chart
    app.kubernetes.io/instance: demo-chart
    app.kubernetes.io/version: "1.16.0"
    app.kubernetes.io/managed-by: Helm
  annotations:
    "helm.sh/hook": test
spec:
  containers:
    - name: wget
      image: busybox
      command: ['wget']
      args: ['demo-chart-webapp-chart:8080']
  restartPolicy: Never
```

helm template demo-chart ./ --debug

```
lly@lly ~/m/e/t/2/webapp-chart (master) [1]> helm upgrade webapp-release ./ --set extraLab
els.labels.semester="autumn"
Release "webapp-release" has been upgraded. Happy Helming!
NAME: webapp-release
LAST DEPLOYED: Sun Apr 12 23:31:18 2026
NAMESPACE: default
STATUS: deployed
REVISION: 2
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace default -l "app.kubernetes.io/name=webapp
-chart,app.kubernetes.io/instance=webapp-release" -o jsonpath="{.items[0].metadata.name}")
  export CONTAINER_PORT=$(kubectl get pod --namespace default $POD_NAME -o jsonpath="{.spe
c.containers[0].ports[0].containerPort}")
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl --namespace default port-forward $POD_NAME 8080:$CONTAINER_PORT
```

helm upgrade webapp-release ./ --set extraLabels.labels.semester=”autumn”

```
Labels:                app.kubernetes.io/instance=webapp-release
                       app.kubernetes.io/managed-by=Helm
                       app.kubernetes.io/name=webapp-chart
                       app.kubernetes.io/version=1.16.0
                       course=kubernetes
                       environment=production
                       helm.sh/chart=webapp-chart-0.1.0
                       project=university
                       semester=autumn
                       team=backend
```

```
lly@lly ~/m/e/t/2/webapp-chart (master)> kubectl get pods -l project=university
NAME                                          READY   STATUS    RESTARTS   AGE
webapp-release-webapp-chart-65b58f7b86-9c2ch   1/1     Running   0          116s
webapp-release-webapp-chart-65b58f7b86-9rqgs   1/1     Running   0          118s
webapp-release-webapp-chart-65b58f7b86-sbxs9   1/1     Running   0          115s
```

kubectl get pods -l project=university

```
lly@lly ~/m/e/t/2/webapp-chart (master)> kubectl exec webapp-release-webapp-chart-65b58f7b
86-9c2ch -- printenv | grep APP_
APP_ENV=production
APP_DEBUG=false
WEBAPP_RELEASE_WEBAPP_CHART_SERVICE_PORT_HTTP=8080
WEBAPP_RELEASE_WEBAPP_CHART_PORT_8080_TCP_PROTO=tcp
WEBAPP_RELEASE_WEBAPP_CHART_PORT_8080_TCP_PORT=8080
WEBAPP_RELEASE_WEBAPP_CHART_SERVICE_HOST=10.110.59.248
WEBAPP_RELEASE_WEBAPP_CHART_PORT_8080_TCP=tcp://10.110.59.248:8080
WEBAPP_RELEASE_WEBAPP_CHART_SERVICE_PORT=8080
WEBAPP_RELEASE_WEBAPP_CHART_PORT=tcp://10.110.59.248:8080
WEBAPP_RELEASE_WEBAPP_CHART_PORT_8080_TCP_ADDR=10.110.59.248
```

kubectl exec <pod_name> -- printenv | grep APP_

Задание 3

```
lly@lly ~/m/e/t/2/webapp-chart (master)> helm install webapp-dev ./ -f values-development.
yaml
NAME: webapp-dev
LAST DEPLOYED: Sun Apr 12 23:50:45 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace default -l "app.kubernetes.io/name=webapp
-chart,app.kubernetes.io/instance=webapp-dev" -o jsonpath="{.items[0].metadata.name}")
  export CONTAINER_PORT=$(kubectl get pod --namespace default $POD_NAME -o jsonpath="{.spe
c.containers[0].ports[0].containerPort}")
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl --namespace default port-forward $POD_NAME 8080:$CONTAINER_PORT
```

helm install webapp-dev ./ -f values-development.yaml

```
lly@lly ~/m/e/t/2/webapp-chart (master)> kubectl get configmaps
NAME                           DATA   AGE
kube-root-ca.crt               1      18d
webapp-dev-webapp-chart-config 3      29s
```

kubectl get configmaps

```
lly@lly ~/m/e/t/2/webapp-chart (master)> kubectl describe configmap webapp-dev-webapp-char
t-config
Name:         webapp-dev-webapp-chart-config
Namespace:    default
Labels:       app.kubernetes.io/managed-by=Helm
Annotations:  meta.helm.sh/release-name: webapp-dev
              meta.helm.sh/release-namespace: default

Data
====
APP_DEBUG:
----
true

APP_ENV:
----
development

LOG_LEVEL:
----
debug


BinaryData
====

Events:  <none>
```

kubectl describe configmap webapp-dev-webapp-chart-config

```
lly@lly ~/m/e/t/2/webapp-chart (master)> kubectl get deployments -l environment=production
NAME                      READY   UP-TO-DATE   AVAILABLE   AGE
webapp-prod-webapp-chart  3/3     3            3           33s
```

kubectl get deployments -l environment=production

Задание 4

```
lly@lly ~/m/e/t/3/problematic-chart (master)> helm lint
==> Linting .
[INFO] Chart.yaml: icon is recommended
[ERROR] templates/: parse error at (problematic-chart/templates/deployment.yaml:24): unexp
ected "}" in end

Error: 1 chart(s) linted, 1 chart(s) failed
```

helm lint

```
lly@lly ~/m/e/t/3/problematic-chart (master)> helm template problematic-chart ./ --debug
install.go:225: 2026-04-12 23:57:05.57983267 +0300 MSK m=+0.020241292 [debug] Original cha
rt version: ""
install.go:242: 2026-04-12 23:57:05.57986918 +0300 MSK m=+0.020277792 [debug] CHART PATH:
/home/lly/ma/eleventh/tasks/3/problematic-chart


Error: parse error at (problematic-chart/templates/deployment.yaml:24): unexpected "}" in
end
helm.go:92: 2026-04-12 23:57:05.580948231 +0300 MSK m=+0.021356853 [debug] parse error at
(problematic-chart/templates/deployment.yaml:24): unexpected "}" in end
```

helhelm template problematic-chart ./ --debug

```
lly@lly ~/m/e/t/3/problematic-chart (master) [1]> helm install test-release ./ --dry-run
Error: INSTALLATION FAILED: parse error at (problematic-chart/templates/deployment.yaml:24
): unexpected "}" in end
```

helm install test-release ./ --dry-run

```
lly@lly ~/m/e/t/3/problematic-chart (master)> helm install fixed-release ./
NAME: fixed-release
LAST DEPLOYED: Sun Apr 12 23:59:18 2026
NAMESPACE: default
STATUS: deployed
REVISION: 1
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace default -l "app.kubernetes.io/name=proble
matic-chart,app.kubernetes.io/instance=fixed-release" -o jsonpath="{.items[0].metadata.nam
e}")
  export CONTAINER_PORT=$(kubectl get pod --namespace default $POD_NAME -o jsonpath="{.spe
c.containers[0].ports[0].containerPort}")
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl --namespace default port-forward $POD_NAME 8080:$CONTAINER_PORT
```

helm install fixed-release ./

```
lly@lly ~/m/e/t/3/problematic-chart (master)> kubectl get pods -l app.kubernetes.io/instan
ce=fixed-release
NAME                                          READY   STATUS    RESTARTS   AGE
fixed-release-problematic-chart-8dd6df6d9-7t74p   1/1     Running   0          42s
```

kubectl get pods -l app.kubernetes.io/instance=fixed-release

Контрольные вопросы

В чем основное преимущество Helm перед прямым применением kubectl apply? – имеет шаблоны, управляет всем приложением

Какие основные файлы входят в структуру Helm Chart? – deployment.yaml, service.yaml, values.yaml

Как работает шаблонизация в Helm? – используются go templates, значения пишутся в {{ }}

В чем разница между Chart и Release? – chart – это шаблон, а Release – установленный экземпляр

Как можно передавать пользовательские значения при установке Chart? – через файл или –set

Что такое зависимости в Helm? – другие Chart, которые подключаются к основному

Как можно отладить Helm Chart перед установкой? – lint, template, install –dru-run –debug

Какие команды используются для управления жизненным циклом Helm Release? – install, upgrade, delete, list